Ahnaf Samin

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RESEARCH INTERESTS

Deep learning for critical-infrastructure security, including time-series anomaly and intrusion detection; natural language processing for low-resource languages (Bengali); large language models and agentic AI systems.

EDUCATION

Ahsanullah University of Science and Technology Feb 2020 – Apr 2024
B.Sc. in Computer Science and Engineering • CGPA: 3.48 / 4.00 Dhaka, Bangladesh
Thesis: *Emoji-infused Bengali sarcasm and offensive language analysis: A novel approach to context-aware text understanding* (published at ICCIT 2025) • Supervisor: Ms. Tahsin Aziz

Dhaka Residential Model College 2017 – 2019
Higher Secondary Certificate (Science) • GPA: 5.00 / 5.00 Dhaka, Bangladesh

Rani Bilashmoni Government Boys' High School 2017
Secondary School Certificate • GPA: 5.00 / 5.00 Gazipur, Bangladesh

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES

1. **Samin, A.**, Zahin, F., Duti, N. I., Chowdhury, B. S., Akand, A. R., Arman, M., Das, R., & Zuberi, N. (2026). *Hybrid deep learning for smart grid SCADA intrusion detection with mutual information feature selection for smart meter data analysis.* Scientific Reports. Published online 14 Sep 2026. doi:10.1038/s41598-026-70577-4
   First-authored. A hybrid CNN–Transformer–BiGRU model for intrusion detection on SCADA-based smart-meter data, using mutual-information feature selection and ADASYN class balancing; achieves 0.94 ROC-AUC on the MSU power-system benchmark.
2. Arman, M., **Samin, A.**, Islam, A. K. M. M., Arnob, M. M. R., Islam, M. J., & Mamoon, I. A. (2026). *PruDensNet: A parameter-efficient depthwise separable CNN for MRI-based brain tumor classification.* Frontiers in Medicine, 13, 1743015. doi:10.3389/fmed.2026.1743015
   Conceived the PruDensNet architecture and led the research. A lightweight (≈1.46 M-parameter) attention-augmented CNN reaching 96.05% test accuracy on four-class brain-tumor MRI under a leakage-aware evaluation protocol.

PEER-REVIEWED CONFERENCE PAPERS

1. **Samin, A.**, Arnob, M. M. R., Anjum, J., Khan, M. N. Y., Nafi, I. R., & Aziz, T. (2025). *Emoji-infused Bengali sarcasm and offensive language analysis: A novel approach to context-aware text understanding.* 28th International Conference on Computer and Information Technology (ICCIT 2025), IEEE.
   First-authored; undergraduate thesis. Introduced **AUST-Sarc**, a publicly released emoji-aware Bengali sarcasm/offensiveness corpus, and benchmarked transformer-based detection on noisy, code-mixed social text.

DATASETS & RESEARCH ARTIFACTS

AUST-Sarc: Emoji-Aware Bangla Sarcasm & Offensiveness Dataset 2025
2,649 emoji-bearing Bangla sentences labeled for sarcasm and offensiveness • Mendeley Data • doi:10.17632/7ryvn5gw88.2

SELECTED RESEARCH & ENGINEERING PROJECTS

RAISE-Me: Responsible Agentic Intelligence for Skills and Employment Technical Project Lead, D-Ready
Production multi-agent AI platform spanning skills assessment to employment

- Architected a four-agent system: **Evaluate-Me** (CV parsing and skills-gap analysis), **Train-Me** (personalised AI tutor with rubric-based assessment), **Jobify-Me** (orchestration and agent handoffs), and **Place-Me** (employer matching).
- Built multi-agent orchestration across OpenAI, Anthropic, and Gemini models, with RAG over vector databases and real-time voice tutoring.
- Embedded responsible-AI governance (explainability, bias detection, human-in-the-loop oversight, audit trails) aligned to ISO/IEC 42001 and the EU AI Act.

Real-Time Bangla Sign Language Translation Academic
Deep learning with holistic body/hand keypoints for sentence-level Bangla sign translation

- Ranked **1st of 20 projects** in the CSE 4238 Soft Computing Lab; awarded an A+ grade.

ASR Evaluation System GitHub
Benchmarking framework for Bangla and English automatic speech recognition

- Implemented WER/CER scoring, text normalization, and reusable evaluation modules for comparing pretrained ASR models.

Autonomous Trading Agentic AI (MVP) GitHub
Agentic LLM workflow with orchestration, tool use, and explainable reasoning

- Built a modular architecture covering tool invocation, state handling, guardrails, and rule-constrained decisions over structured market data.

PROFESSIONAL EXPERIENCE

Digital Readiness & Intelligent Systems Ltd. (D-Ready) Apr 2026 – Present
AI Engineer • Remote (UK-based)

- Sole developer and project lead of RAISE-Me (see Projects), taken from concept to a working multi-agent product.

Huawei Technologies (Bangladesh) Ltd. Jul 2024 – Apr 2026
System Engineer, Analytics & AI Automation Dhaka, Bangladesh

- Led the PowerStar 2.0 energy-optimization deployment across Robi Axiata's nationwide network, achieving **7.79% network-wide power savings** from pilot through permanent rollout.
- Applied ML models for KPI forecasting and anomaly detection on telecom network data, enabling proactive efficiency decisions.
- Built Python and SQL ETL pipelines integrating LTE and GSM datasets for automated KPI analysis and reporting.

TECHNICAL SKILLS

Languages	Python (primary), TypeScript, JavaScript, Java, C/C++, SQL
ML & Deep Learning	PyTorch, TensorFlow, Keras, scikit-learn; CNNs, RNNs, transformers, transfer learning
LLMs & Agents	Hugging Face Transformers, LangChain; RAG, embeddings, vector databases; multi-agent systems, tool calling, LLM evaluation; OpenAI, Anthropic, and Gemini APIs
Data	pandas, NumPy, Matplotlib, seaborn; PostgreSQL, MySQL, MongoDB; FAISS, ChromaDB
Tools	Git/GitHub, Docker, GCP, FastAPI, LaTeX

AWARDS & HONORS

- 4× **Galloping Horse Award**, Huawei Technologies (2024–2026): PowerStar 2.0 Project Leader; CX Swap Dashboard; Best Insight Presentation; SEC Rising Star.
- **1st Place (of 20)**, Soft Computing Lab project showcase, AUST (Bangla Sign Language Translation).

TEST SCORES & LEADERSHIP

- **IELTS**: Band 7.0. **General Secretary**, AUST Debating Club (2023–2024).

REFERENCES

Mr. Chandra Shekhar Shill
SEC Director, South Asia Rep. Office
Huawei Technologies (Bangladesh) Ltd.
chando.shager@huawei.com

Ms. Tahsin Aziz
Former Assistant Professor, CSE, AUST
Thesis Supervisor
tahsinaziz12.cse@gmail.com

Ms. Syeda Shabnam Hasan
Assistant Professor, Dept. of CSE, AUST
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Ms. Tanjila Broti
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